

Section B: K-Coloring Solution

Problem: K-Coloring

Definition:

K-Coloring asks whether graph vertices can be colored using at most k colors such that adjacent vertices have different colors.

Task 1 Solution:

A bipartite graph has chromatic number:
2 (unless it has no edges, then it is 1).

A complete graph K_n has chromatic number:
 n .

Task 2: K-Coloring is in NP

Certificate:

A coloring assignment for every vertex.

Verification:

For every edge (u,v) , check:

$\text{color}(u) \neq \text{color}(v)$

This requires $O(E)$ time.

Task 3 and 4: Reduction from 3-Coloring

Transform a 3-coloring instance into a k -coloring instance.

Keep the same graph and set $k=3$ for the base case.

For larger k , add forced-color structures so additional colors cannot change the original satisfiability.

The transformation is polynomial.

Correctness:

A valid 3-coloring gives a valid k -coloring.

A valid k -coloring of the transformed graph gives a valid 3-coloring of the original graph.